

Chapter 1 & 2:

Introduction to Regression Models and Basic Statistical Concepts

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October 4, 2025

Conceptual Summary (Core Ideas)

1. Population Regression Function (PRF)

The *theoretical* relationship between the dependent variable Y and the explanatory variable X :

$$E(Y | X) = \beta_1 + \beta_2 X$$

This represents the *expected* or *average value* of Y given a specific value of X .

2. Sample Regression Function (SRF)

Since we don't have access to the entire population, we estimate the parameters β_1, β_2 using sample data:

$$\hat{Y}_i = \hat{\beta}_1 + \hat{\beta}_2 X_i$$

3. Covariance and Correlation

- *Covariance* shows whether X and Y move together (positive or negative).
- *Correlation coefficient* (r) standardizes covariance and lies between -1 and $+1$:

$$r = \frac{Cov(X, Y)}{\sigma_X \sigma_Y}$$

4. Interpretation of $E(Y | X)$

It is the mean (expected value) of Y for a given value of X . As X changes, the expected value of Y also changes.

5. Correlation vs. Causation

Correlation shows association, *not* causality.

Example: Ice cream sales and drowning cases are correlated in summer, but one does not cause the other.

Questions

1. Why is $E(Y | X)$ called a *conditional mean*?
2. What does the correlation coefficient tell us about two variables?
3. If $Cov(X, Y) = 0$, does that mean X and Y are independent?
4. What's the difference between the PRF and the SRF?
5. Why do we use the regression function for prediction?

Problem Set

Problem 2.1 - Conditional Expectation

Suppose the following joint probability distribution between X (years of education) and Y (monthly income in hundreds of dollars) is given:

X (years)	Y (hundreds of \$)	P(X,Y)
10	15	0.1
10	20	0.1
12	25	0.2
12	30	0.2
16	40	0.2
16	45	0.2

Tasks:

1. Compute $E(Y | X = 10)$, $E(Y | X = 12)$, and $E(Y | X = 16)$.
2. Derive the *population regression function (PRF)*: $E(Y | X) = \beta_1 + \beta_2 X$.
3. Interpret β_2 .

Problem 2.2 – Covariance and Correlation

You observe the following data:

X	Y
1	2
2	3
3	4
4	6

Tasks:

1. Calculate means $\bar{X}$ and $\bar{Y}$.
2. Compute the covariance $Cov(X, Y)$.
3. Compute the correlation coefficient r .
4. Interpret whether the relationship is strong, weak, positive, or negative.

Problem 2.3 – Interpreting Correlation vs. Causality

A researcher finds that ice cream sales and drowning incidents have $r = 0.85$.

Tasks:

1. Explain why this does not imply causality.
2. Suggest what hidden variable might explain this correlation.